(d) With the help of neat block diagram describe the system arrangement in an intensive care unit. 7

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B. E. (Sixth Semester) Examination, Nov.-Dec. 2015

(Old Scheme)

(EEE, Et & T Branch)

MEDICAL ELECTRONICS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Attempt all questions. Part (a) is compulsory. Attempt any two parts from (b), (c) & (d) parts.*

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Unit-I

1. (a) How a cell can be depolarized? 2

(b) Mention the component of man-instrument system and also describe their functions. 7

(c) Explain in detail the various physiological system of body. 7

(d) Explain various parameters associated with respiratory system. 7

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Unit-II

2. (a) Signal conditioning play important role in biomedical field. Justify. 2

(b) Define Bio-electric potential with suitable wave form. Also explain the resting potential and action potential. 7

(c) Explain the working of an ECG recorder with emphasis to various lead configurations. 7

(d) Draw the block diagram of EEG and explain each block in details. 7

Unit-III

3. (a) How is ionizing radiation different from other electromagnetic radiation. 2

(b) Explain phonocardiograph. 7

(c) Explain the various methods of blood pressure measurement. 7

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(d) Why ultracound is important in medical diagnosis system. Discuss the principles of operation of medical diagnosis using ultrasound. 7

Unit-IV

4. (a) Explain the limitations of patient monitoring system. 2

(b) Define pacemaker. Draw block diagram of internal pacemaker and explain. 7

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(c) A defibrillator delivers a square pulse of 4000 V with a duration 4 mili seconds. The internal resistance of the defibrillator is about 15 Ω. The skin electrode resistance is 50 Ω and thorax resistance is 30 ohms. Compute the eneergy delivered to the patient's throax and the total energy available from the detibrillator. Also calculate the percentage of loss of energy? 7

(d) Mention the effect of electrical shock to human body. How patient electrical safety is taken care of in a hospital. 7

Unit-V

5. (a) Define bio-telemetry. 2

(b) Discuss the problems encountered in implont bio-telemetry. Mention the uses of biotelemetry. 7

(c) What are the various biological parameter that are monitored by a patient monitoring system. 7

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